Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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Abstract

Background. Model and preprocessing choices alter the accuracy, cost and cross-cohort errors of automatic sleep staging, but are rarely tested on data unused during development.

Methods. Six single-channel EEG configurations were trained on 78 Sleep-EDF subjects with shared subject-wise ten-fold partitions. Zero-shot transfer was examined on 180 Sleep Heart Health Study Visit 1 (SHHS1) participants and tested on 500 further participants under a protocol committed beforehand.

Results. A one-dimensional residual network with a temporal convolutional network (ResNet-1D-TCN) on band-pass-filtered input (E3) reached a pooled Sleep-EDF macro-F1 of 0.7904 and, despite 4.37× as many parameters as a CNN-BiLSTM baseline (E0), ran 3.76× faster at inference and trained 10.2× faster, because E0 needed 22.5× as many optimiser steps. On the 500 further participants, E3 improved subject-mean macro-F1 over E0 by 0.0467 ($CI_{95\%}$ 0.0403–0.0531), over record-wise z-scoring (E6) by 0.0293 (0.0231–0.0354) and over unfiltered input (E2) by 0.0515 (0.0460–0.0571), confirming all three pre-specified hypotheses. Across the evaluated seeds, band-pass-only input (E4) exceeded E0 and E2 by 0.031–0.070, more than E4's between-seed range (0.019), and 90-minute rolling z-scoring, despite matching input scale across cohorts, still scored below E4. E0 and E3 classified more than 72% of N3 (deep-sleep) epochs as N2, although a Sleep-EDF amplitude threshold separated 84% of them from N2; neither rescaling nor domain adaptation removed this error without losses elsewhere.

Conclusions. Band-pass filtering accounted for most of the cross-cohort gain but did not resolve N3-to-N2 confusion. Pipeline comparisons should report variation between training runs and be confirmed on held-out data.

Keywords: sleep staging; single-channel EEG; temporal convolutional network; preprocessing; domain shift; external validation.

1. Introduction

Sleep staging assigns one of five labels, namely W, N1, N2, N3 and REM, to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained staff and is not fully reproducible: in the inter-scorer reliability programme of the American Academy of Sleep Medicine, agreement is lowest for N1 [1]. An automated system therefore has to be judged not by overall accuracy alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Published results in this field are hard to compare. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, which makes architectural effects difficult to separate from other pipeline differences. For practical development, what matters is which pipeline choices improve prediction, what they cost computationally, and whether their benefits persist on another cohort. Answering these questions requires shared evaluation partitions, an explicit account of what changes between configurations, and evaluation data that did not influence the choices being compared.

We take ZleepAnlystNet [2], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal baseline, E0, reimplemented under the present protocol. E1 replaces the BiLSTM with a temporal convolutional network (TCN) [3], reusing the encoder and cached features but changing the sequence-model training recipe. E2 replaces E1's 75 features, which describe the current, previous and next epochs (C/P/N), with 128 features from a residual encoder [4] that sees only the current epoch, and also changes encoder training and selection. Each step therefore compares complete configurations rather than isolating a single architectural change. Preprocessing contrasts then examine filtering and amplitude handling within the same ResNet-1D-TCN model and training configuration. The study does not propose a new architecture: its contribution is a controlled comparison of existing components whose main findings are then tested on participants held out from development.

We ask three questions:

(i) Which of the evaluated changes, to the sequence model, the epoch encoder, signal filtering or amplitude handling, account for differences in accuracy, computational cost and transfer to another cohort?

(ii) Do the resulting differences between pipelines hold on participants who played no part in development, and are they larger than the variation between training runs?

(iii) Which sleep-stage errors dominate after transfer, how do they depend on input amplitude, and can prior correction, class weighting or domain adaptation reduce them without losses in other stages?

Our contributions follow these questions.

(1) We show that the implemented ResNet-1D-TCN pipeline runs three to four times faster than the CNN-BiLSTM baseline despite having more parameters, and needs about one tenth of the training time because it trains two networks per fold instead of sixteen. Among the evaluated changes, band-pass filtering rather than the encoder or sequence model accounts for most of its cross-cohort advantage: without filtering, the same model performed close to the baseline.

(2) We confirm this advantage on 500 held-out SHHS1 participants under a protocol fixed before their data were processed. Across five seeds of the band-pass-only variant and two seeds of each comparator, its contrasts with the baseline and the unfiltered model exceed the observed between-seed range, whereas the band-pass-only and fixed-scale variants, whose input-scale difference batch normalisation largely absorbs, differ by less than that range. Confidence intervals computed for fixed models capture participant sampling but not retraining.

(3) We identify N3-to-N2 confusion as a shared cross-cohort error that recurs in every zero-shot pipeline and in the held-out sample. An input-amplitude sweep shows that this error depends on, but is not explained by, signal amplitude, and a stage-wise description of the input shows that most SHHS1 N3 epochs remain separable from N2 by an amplitude threshold chosen on Sleep-EDF. In matched experiments, none of prior correction, class weighting, batch-normalisation adaptation or adversarial domain adaptation reduced this error without losses in other stages; the failure of prior correction persisted with priors estimated from 500 recordings.

2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [5]. AttnSleep introduced attention over multi-resolution features [6], and SleepTransformer applied self-attention with explicit uncertainty quantification [7]. Sequence-to-sequence, compact and multi-view models followed [8, 9, 10], and more recent work extends the modelled context to whole sleep cycles or adds contrastive pre-training [11, 12]. Together these establish that sequential context matters. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in a recent review [13].

Single-channel systems also differ in how they represent the signal. SleepInceptionNet systematically compared raw, spectral and time-frequency inputs and selected a continuous-wavelet, central-channel Inception model that scores one epoch at a time [14]. That study supports two points relevant here: a single EEG channel can support five-stage scoring, and the choice of input representation is part of the model definition rather than a neutral implementation detail. Van Der Donckt et al. [15] further showed that increasingly complex deep architectures need not outperform a carefully designed conventional pipeline under a matched protocol, and a feature-based classifier trained on several cohorts generalised to unseen recordings [16].

Cross-dataset sleep staging is an active methodological problem. Models trained on many cohorts and channel configurations, such as U-Sleep [17] and the mixed-cohort residual networks of Olesen et al. [18], seek robustness through the diversity of their training data; RobustSleepNet accepts arbitrary montages for the same purpose [19], and U-Sleep tolerates derivations outside the scoring guidelines [20]. Other methods instead adapt to the target domain, by fine-tuning on a small labelled target sample [21], by adversarial alignment with few or no target labels [22], or by aligning features across several source domains so that no target data are needed [23]. He et al. [24] trained a transfer-learning network across Sleep-EDF, SHHS1 and PhysioNet Challenge data, and ADAST uses domain-specific attention, adversarial alignment and iterative pseudo-label self-training on unlabelled target data [25]. Our study addresses a narrower question that these approaches leave open: within a single-source pipeline, which model and preprocessing choices determine transfer, what they cost, and whether their effects hold on data that did not inform them.

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [26, 27, 28]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [29], distributed through the National Sleep Research Resource (NSRR) [30]. The two differ in age, comorbidity, recording hardware, montage and label distribution, so SHHS tests a form of cohort change not represented by a held-out split of Sleep-EDF. We use it to assess robustness to such a change; a retrospective transfer test is not a clinical validation.

Two further strands of work inform the interpretation. Slow oscillations have a spatially varying amplitude and propagate across the cortex [31], while deep-sleep scoring uses amplitude criteria [32, 33]. A recent SHHS analysis examined how those criteria relate to age- and sex-associated differences in N3 scoring [34]. These studies give context for transfer between derivations and populations; the present experiment does not establish such a mechanism. Differences in class prevalence can also affect predictions; prior correction and probability calibration address related but distinct problems [35, 36].

3. Materials and methods

3.1. Data and labels

The Sleep Cassette subset of Sleep-EDF Expanded used here comprises 153 recordings from 78 subjects; we use the Fpz-Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM, merging the original stages 3 and 4 into N3; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window, applied identically to both cohorts, extends from 30 min before the first to 30 min after the last epoch scored N1, N2, N3 or REM. Because it strongly affects the wake fraction and every

Table 1. Scored-label distribution in the three evaluation samples. The ratio compares the share in the holdout SHHS1 sample with that in Sleep-EDF.

	Sleep-EDF		**SHHS1 first** (n = 180)		**SHHS1 holdout** (n = 500)		
Stage	**Epochs**	**Share**	**Epochs**	**Share**	**Epochs**	**Share**	**Ratio**
W	65,941	33.73%	36,983	21.88%	99,674	21.74%	0.64
N1	21,522	11.01%	7,002	4.14%	18,930	4.13%	0.38
N2	69,132	35.37%	76,287	45.14%	207,848	45.34%	1.28
N3	13,039	6.67%	22,806	13.49%	61,523	13.42%	2.01
REM	25,835	13.22%	25,934	15.34%	70,434	15.36%	1.16
Total	**195,469**	**100%**	**169,012**	**100%**	**458,409**	**100%**	

aggregate metric, and because its anchors use reference labels, it is a benchmark convention rather than a deployable inference rule.

The SHHS Visit-1 selection admitted values 3–7 of the quality field `overall_shhs1`, which all 5,804 participants in the dataset release met. Participant identifiers were ranked by SHA-256 with selection seed 42, and the first 220 were assigned roles: 5 adaptation, 15 validation, 180 test and 20 reserve participants. The validation group served as a technical gate: the complete inference run had to finish without errors on it before the test participants were processed, and no model or setting was chosen on it. The adaptation group provided the five unlabelled recordings used for prior estimation and by ADAST. We call the 180 test participants (age 63.3 ± 11.5 years; 93 men and 87 women) the *first SHHS1 sample*. It was selected before any prediction was reviewed, but all subsequent analyses and interventions were developed on it.

To test the main findings on data that had not informed them, we drew a *holdout SHHS1 sample* of the participants ranked 221 to 720 in the same ranking; one participant without an available recording was replaced by the next-ranked participant before any processing. The 500 holdout participants (age 62.8 ± 11.5 years; 249 men and 251 women; recordings of 5.5–9.1 h, median 8.8 h) had not been used for any earlier step of the study, and all passed the technical audit. The sample size was set by the cost of downloading and processing recordings rather than by a power calculation; the confirmatory intervals had half-widths of about 0.006.

For both samples the EEG channel is C4-A1 at 125 Hz, resampled to 100 Hz with a polyphase anti-aliasing filter (up 4, down 5, Kaiser window $\beta = 5.0$) applied to the continuous signal before epoching. SHHS labels are mapped 0 →W, 1 →N1, 2 →N2, $\{3, 4\}$ →N3, 5 →REM, $\{6, 9\}$ →excluded. Table 1 reports the three label distributions.

Both SHHS1 samples contain about twice the N3 share and 0.38 times the N1 share of Sleep-EDF; for the first sample, the total variation distance from Sleep-EDF is 0.187 and $D_{\mathrm{KL}}(\mathrm{EDF} \,\|\, \mathrm{SHHS})$ is 0.101 nats.

3.2. Preprocessing conditions

Figure 1 summarises the evaluated architecture, context and preprocessing configurations and their shared evaluation protocol. Calibrated EEG is processed continuously before epoching; the resulting epochs are encoded by either the 15-CNN or ResNet-1D encoder, then modelled as a sequence by a BiLSTM or TCN.

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four active signal conditions enter the performance analysis, and one audit-only condition (E5) records a planned clipping ablation:

- **Raw** (E0–E2): no filtering, no clipping, no rescaling; native μV retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5–30 Hz, implemented as second-order sections with forward-backward filtering.
- **Band-pass plus clipping** (E5; audit only): the E4 signal followed by a hard clip at ±800 μV. E5 and E4 inputs were bitwise identical in all 153 Sleep-EDF recordings, so E5 was cancelled after fold 0, before any score was inspected.

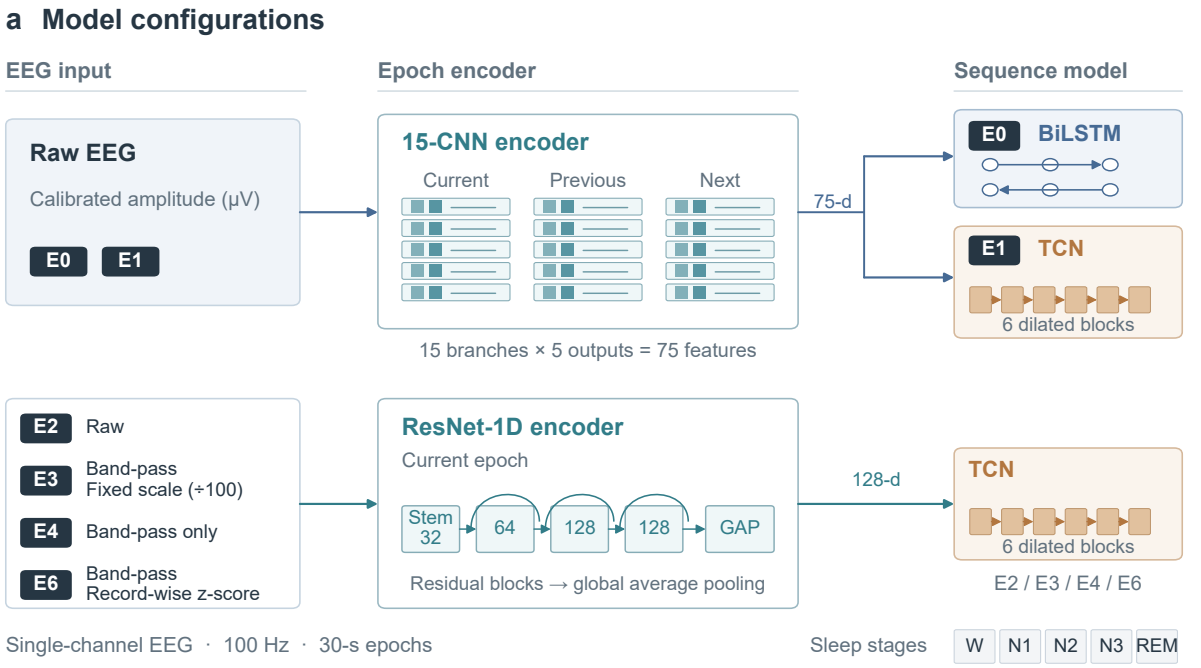


Figure 1. Model configurations and evaluation protocol. (a) E0 and E1 share a 15-CNN encoder: five branches for each of the current, previous and next epochs yield 75 features for a BiLSTM or TCN. E2, E3, E4 and E6 use separately fitted current-epoch ResNet-1D encoders and TCNs. ResNet numbers denote output channels; GAP is global average pooling. Both TCNs have six non-causal dilated blocks. E3 and E6 additionally clip at ±800 μV after band-pass filtering (the clip never activated); E3 divides by 100, whereas E6 uses complete-record z-score statistics. (b) Sleep-EDF uses ten shared subject-wise folds, with eight for training, one for validation and one for testing per rotation. SHHS1 evaluation averages class probabilities from the ten source-fold checkpoints before selecting the predicted class, without target-domain model fitting; the panel lists the configurations evaluated on the holdout sample, and E1 was evaluated on the first sample only. E6 uses unlabelled target-record statistics; E6R (amendment A2, not drawn) replaces them with a 90-minute rolling z-score. E5 is an input-identity audit, not an additional performance configuration. The symbols represent architecture and fold roles, not measured signals or partition sizes.

- **Band-pass, clipping and division by 100** (E3): the same band-pass, followed by a hard clip at ±800 μV and division by a constant factor of 100. No record-level or cohort-level statistic is estimated.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This uses no target labels but does use all samples of the target recording, removes absolute amplitude differences between recordings and is transductive at the recording level.

No notch filter is applied: 50 Hz mains interference in Sleep-EDF lies at the Nyquist frequency of the 100 Hz signal, and 60 Hz interference in the 125 Hz SHHS recordings is attenuated by the anti-aliasing filter of the resampling step; both lie outside the 0.5–30 Hz pass-band.

The ±800 μV clip never activated: band-pass-filtered Sleep-EDF values did not exceed 246 μV, every holdout SHHS1 EEG channel has a physical range of ±125 μV, and the preprocessing audit of the first SHHS1 sample recorded no clipped samples. E3 therefore differs from E4 only by the factor of 100. The bias-free first ResNet-1D convolution is followed by batch normalisation, which largely absorbs a constant

Table 2. Experimental configuration codes. The identifiers denote signal-model combinations, not a performance ranking or a sequence of model versions.

ID	Signal condition	Epoch encoder	Sequence model and role
E0	Raw	15-branch CNN	BiLSTM; internal baseline
E1	Raw	15-branch CNN	TCN; sequence-model and training-recipe replacement
E2	Raw	ResNet-1D	TCN; encoder/context and encoder-training replacement
E3	Band-pass + clip + ÷100	ResNet-1D	TCN; band-pass configuration
E4	Band-pass only	ResNet-1D	TCN; band-pass configuration without scaling
E5	Band-pass + clip	ResNet-1D	TCN; audit only, identical input to E4
E6	Band-pass + clip + z-score	ResNet-1D	TCN; record-wise normalisation
E6R	Band-pass + clip + 90-min z-score	ResNet-1D	TCN; windowed normalisation (amendment A2)

input scale. This invariance is approximate: the finite stabilising constant ε, running-statistic initialisation and numerical precision can affect outputs and training trajectories (Supplementary Section S4). E3−E4 therefore compares separately trained fixed-scale and band-pass-only variants, not mathematically identical runs. E3−E2 compares filtering and fixed scaling together with variation between training runs; the E4 results support the contribution of filtering without the fixed scaling step.

The record-wise statistics of E6 are computed over very different spans in the two cohorts. Sleep-EDF recordings last 17–24 h (median 23 h) and include long daytime segments outside the evaluation window, whereas SHHS1 recordings cover the night in bed. Within the evaluation window, the median standard deviation of E6 inputs was 0.77 in Sleep-EDF but 0.99 in the holdout SHHS1 sample (interquartile range 0.85–1.00). E6 thus presents SHHS1 signals to the model at about 1.3 times the relative scale seen in training, in addition to removing between-recording amplitude differences. A later amendment therefore added E6R, which computes the z-score over a centred 90-minute window so that both cohorts reach the model at the same relative scale (Section 3.10).

3.3. Model configurations

Six configurations, E0–E4 and E6, enter the main performance analysis, and E6R enters the second amendment. The baseline encoder concatenates five-class softmax outputs from 15 CNN branches spanning the current, previous and next epochs (C/P/N), giving 75 features per epoch; ResNet-1D produces 128 features from the current epoch only. The non-causal six-block TCN has a 253-epoch receptive field, and the BiLSTM baseline has 128 hidden units per direction. All models are offline: the sequence models use future context, and zero-phase filtering and E6 normalisation use future samples. Layer dimensions and hyperparameters are given in Supplementary Methods S1.

3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into ten subject-disjoint folds. In each round one fold is the test fold, the next is the validation fold and the remainder are training folds, so both nights of a subject share a role and the test fold never enters model selection; the same partition is used for every configuration. Training has two stages. The epoch encoder uses class weights from the training subjects and is selected on validation data, by minimum weighted loss for each baseline CNN and maximum macro-F1 for ResNet-1D; its checkpoint then caches features without gradients. The sequence model is fitted on the frozen cache with masked cross-entropy and selected by validation macro-F1. E1 reuses the E0 encoder but changes the sequence-optimiser recipe, and E2 also changes encoder optimisation and selection (Supplementary Methods S1), so the comparisons fix partitions and evaluation, not every training choice. Every configuration was trained with seeds 42 and 123,

and E4 additionally with seeds 2025, 2027 and 2029 in the second amendment. On the same Sleep-EDF Fpz-Cz recordings, E0 at seed 42 reached a pooled macro-F1 of 0.7754 and an accuracy of 0.8262, against 0.7826 and 0.8409 reported for ZleepAnlystNet under ten-fold subject-wise cross-validation [2], although E0 was selected on held-out validation subjects rather than on a random epoch-wise split of the training data (Supplementary Table S3).

3.5. Statistical analysis

All configurations completed validation-based selection before the seed-42 test predictions were evaluated. In Sleep-EDF, pooled macro-F1 is the headline metric, but the unit of inference is the subject. A paired subject-cluster bootstrap with 10,000 resamples gives percentile intervals for differences [37], a two-sided Wilcoxon signed-rank test [38] assesses per-subject differences, and Holm correction [39] is applied over the four pre-specified comparisons E1−E0, E2−E1, E3−E2 and E3−E6; E3−E0 is post-hoc. Pooled macro-F1 is computed from summed confusion matrices, whereas subject-mean macro-F1 averages the participants' scores; metrics use five fixed classes and assign zero to an empty denominator. Win/tie/loss counts, sign dominance $(W - L)/(W + L)$ and an exact sign test are descriptive. All intervals are marginal 95% intervals that treat the trained models as fixed and therefore exclude variation between training runs. The seed-123 repeat, completed after the seed-42 results were known, is analysed separately.

For SHHS1, each configuration is applied without weight updates (zero-shot): the softmax probabilities of its ten Sleep-EDF fold models are averaged in float64 before the argmax. On the first SHHS1 sample, the pre-specified primary criterion was subject-mean macro-F1, with E3−E0 and E3−E6 as primary comparisons under the same procedure; comparisons involving E1, E2 and E4 are secondary or post-hoc, and boundary analyses are post-hoc diagnostics (Supplementary Methods S1).

3.6. Confirmatory holdout protocol

Before any holdout recording was preprocessed, a protocol fixing the cohort, preprocessing, checkpoints, ensemble rule, metrics, hypotheses and decision rule was published in the public project repository on 10 October 2026, and the analysis script was published there before it was run (commit identifiers and timestamps in Supplementary Section S3). The confirmatory family comprised three seed-42 contrasts of subject-mean macro-F1, each with an expected positive sign: H1, E3−E0; H2, E3−E6; and H3, E3−E2. A hypothesis was confirmed if its Holm-adjusted two-sided Wilcoxon p-value was below 0.05 and its paired subject-cluster bootstrap interval (10,000 resamples, seed 2040) lay entirely above zero.

The protocol also specified secondary analyses: the seed-123 replication of H1–H3, E4−E3 together with the between-seed differences of E0, E3, E4 and E6, ADAST against its matched control, N3 recall and N3→N2 rate, a present-class variant of subject macro-F1 and the E6 input-scale summary. Twelve arms were evaluated, E0, E2, E3, E4 and E6 at both seeds plus ADAST and its control, with the locked checkpoints on CPU, and no metric was computed before all predictions existed. Deviations, none of which changed a prediction, are listed in Supplementary Section S3.

3.7. Post-confirmatory amplitude and adaptation analyses

After the confirmatory analysis, an amendment published in the repository before its inference specified two descriptive analyses of E4 at seed 42, the band-pass-only form of the pipeline (Section 4.4), on the same 500 participants. First, the input was multiplied by 0.5, 0.75, 1.5 or 2 while the encoder kept its stored batch-normalisation statistics, which tests how strongly the predicted stage depends on signal amplitude. The amendment stated that, if N3 under-detection arises mainly because SHHS1 slow waves reach the encoder at a smaller amplitude than the training data, N3 recall should rise and the N3→N2 rate fall for factors above 1, and that a non-monotonic response would argue against an amplitude-only explanation. Second, adaptive batch normalisation (AdaBN) [40] replaced the running statistics of the ten batch-normalisation layers of the encoder, layer by layer, with those of 20,000 unlabelled SHHS1 epochs; it is a label-free adaptation baseline that updates no weights. We also report the per-recording standard deviation of the filtered input in both cohorts. Conditions were compared with unmodified E4 by paired subject-cluster bootstrap (10,000 resamples, seed 2042) without hypothesis tests. They were computed on a GPU rather than the CPU named in the amendment; GPU and CPU probabilities differed by at most 1.4×10^{-5} with identical predicted stages

(Supplementary Section S5).

3.8. Intervention experiments

After characterising the errors on the first SHHS1 sample, we evaluated three interventions on that sample, each against a matched control and with ten-fold probability averaging: temperature scaling with EM prior correction for the ten seed-123 E3 models, with priors estimated on the five unlabelled adaptation recordings; class-weighted training of the TCN on the frozen seed-123 E3 encoder; and ADAST [25], which adapts to unlabelled target recordings, against a source-only model of the same architecture. ADAST was trained at two learning budgets, 1,140 updates per model and 30 complete source-training passes (full-source training), and evaluated at epoch 30. Model selection used source validation data, and target labels were used only for scoring. The full-source pair was also applied, without further training, to the holdout sample. Supplementary Methods S2 specifies fitting, selection, budgets and development experiments.

3.9. Operational evaluation

Inference timing was measured on one Tesla V100 for all configurations, excluding I/O, preprocessing and training: a batch of one synthetic 100-epoch sequence, 20 warm-up passes and three rounds of 100 timed passes in shuffled configuration order, summarised by the median of all 300 passes. This benchmark measures the implemented forward pipeline for one tensor shape, not algorithmic complexity. Whole-night inference on a CPU was timed post hoc with the inference function of the holdout evaluation: one fold model of E0 and one of E3, which shares its architecture with E2, E4, E6 and E6R, scored the holdout recording of median length (932 epochs, 7.8 h) on a laptop processor (Intel Core i5-13500H, PyTorch 2.5.1) with one and with four threads. After one warm-up run, five runs per model in alternating order were timed, excluding I/O, preprocessing and model loading, and their median is reported (Supplementary Section S9). Training cost was recorded as the wall-clock time of training and validation across all ten folds in the seed-123 run, in which every configuration was trained sequentially on the same Tesla V100 with its own recipe and stopping rule (Supplementary Methods S1).

3.10. Further descriptive analyses

A second amendment (A2), published in the repository before any of its preprocessing, training, inference or analysis, specified five descriptive analyses, with all arms, seeds, settings and the rule by which each result would be reported (Supplementary Section S6). It makes no hypothesis tests and does not revisit H1–H3. (A) E6R repeats E6 with a centred 90-minute rolling z-score at seeds 42 and 123, testing whether the deficit of E6 depended on the span of its statistics. (B) E4 was trained with three further seeds, giving five, to describe training variability. (C) Every fold model of each zero-shot arm was also scored alone on the holdout sample, because Sleep-EDF is evaluated with one model per subject but SHHS1 with ten-model ensembles. (D) The EM prior correction was repeated on the holdout sample with the same ten seed-123 E3 models and temperatures, changing only the data from which the target priors were estimated: random subsets of 5, 20 and 100 recordings (200 draws each), the 20,000 unlabelled epochs of the AdaBN sample, all 500 recordings, and each recording separately. (E) The training time of the seed-123 run was decomposed into the number of optimiser steps, the time per step measured on a Tesla V100, and the remainder. The 50 new Sleep-EDF runs used the locked split, runner and recipes, and their test folds were opened once, after all runs had finished. Intervals are paired subject-cluster bootstrap intervals (10,000 resamples, seed 2043).

Finally, a post-hoc description outside both amendments compared the power spectra of the unfiltered and band-pass inputs in the 153 Sleep-EDF and the 500 holdout recordings (Welch estimate with 20 s Hann segments and 50% overlap, averaged over the valid epochs of each evaluation window). The spectra involve no model and no stage labels beyond the evaluation window (Supplementary Section S7). The description was then extended to the reference stages. For every valid epoch of the band-pass input, the standard deviation and the 0.5–4 Hz power were computed, and the separation of reference N3 from reference N2 epochs by each quantity was summarised by the area under the receiver operating characteristic curve (AUC) within each cohort and by applying to SHHS1 the threshold that maximised balanced accuracy on Sleep-EDF. This analysis uses reference stages but no model or prediction (Supplementary Section S8).

4. Results

4.1. Out-of-fold performance on Sleep-EDF

Table 3. Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

Configuration	macro-F1	Accuracy	κ	F1 (N1)
E0	0.7754	0.8262	0.7614	0.5009
E1	0.7802	0.8315	0.7683	0.5115
E2	0.7835	0.8341	0.7716	0.5180
E3	**0.7904**	**0.8363**	**0.7754**	**0.5276**
E4	0.7891	0.8360	0.7746	0.5270
E6	0.7691	0.8230	0.7573	0.5124

E3 scored highest on all four metrics at seed 42 (Table 3); paired comparisons are given in Table 4.

Table 4. Pre-specified paired comparisons on Sleep-EDF macro-F1 (n = 78 subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3−E0 is post-hoc and is excluded from the Holm family.

Comparison	Δmacro-F1	$CI_{95\%}$	p_{Holm}	W/T/L	Δ_{sign}	p_{sign}
E1−E0	0.0048	[−0.0009, 0.0102]	0.1022	51/0/27	+0.308	0.0088
E2−E1	0.0033	[−0.0024, 0.0088]	0.1036	44/0/34	+0.128	0.3082
E3−E2	0.0070	[0.0003, 0.0145]	0.8989	37/0/41	−0.051	0.7343
E3−E6	**0.0213**	**[0.0122, 0.0307]**	**0.0012**	49/0/29	**+0.256**	**0.0308**
E3−E0 (post-hoc)	0.0150	[0.0057, 0.0259]	n/a	49/0/29	+0.256	0.0308

Among the pre-specified contrasts, only E3−E6 has a pooled bootstrap interval above zero together with a Holm-significant Wilcoxon result (p = 0.0012; Table 4). The E1−E0 and E2−E1 intervals include zero, and E3−E2 has a positive pooled interval although E3 improved on E2 in only 37 of 78 subjects (Holm p = 0.8989), which indicates a small pooled gain rather than a consistent subject-level one. The post-hoc E3−E0 difference of 0.0150 reflects the combined change of architecture, preprocessing and training.

Detailed per-class metrics, the Sleep-EDF confusion matrix and both seed-sensitivity tables are provided in Supplementary Tables S8–S11. On Sleep-EDF, N1 is the hardest class for E3, and the largest E3−E6 F1 difference is on N3, at +0.0628. All four pre-specified differences remain positive under seed 123, but only E3−E6 has a fully positive bootstrap interval in both runs; its Holm-adjusted result is significant for seed 42 only and its magnitude falls from 0.0213 to 0.0102. The direction of the effect is thus consistent across seeds, but its statistical support is not.

4.2. Complexity, latency and training cost

Table 5. Full-pipeline forward benchmark on a Tesla V100 (fold 0, seed 42; batch 1, sequence length 100). Throughput is in epochs s^{-1} and peak allocated memory in MiB; I/O and preprocessing are excluded.

ID	Parameters	Median (ms)	Throughput	Peak mem.	Speed-up
E0	248,630	13.4315	7,445	58.81	1.00×
E1	640,950	12.5111	7,993	60.31	1.07×
E2	1,085,578	3.5750	27,972	75.54	3.76×
E3	1,085,578	3.5687	28,021	75.54	3.76×
E6	1,085,578	3.5756	27,967	75.54	3.76×

The ResNet-1D-TCN family ran 3.76× faster than the CNN-BiLSTM baseline in the forward benchmark (Table 5) while carrying 4.37× the parameters and 16.73 MiB more peak memory, so the implementation is faster but not smaller. E1, which keeps the E0 encoder, gains only 1.07×; the larger gain accompanies the replacement of 15 separately executed CNN branches by one ResNet-1D encoder. Whole-night inference on a CPU gave a similar ratio: one fold model scored a 932-epoch night in a median of 4.31 s for E0 and 1.38 s for E3 with one thread (3.13×), and in 1.75 s and 0.55 s with four threads (3.18×; Supplementary Table S57). A ten-model E3 ensemble thus scores a night in about 14 s on one processor thread, and both pipelines are fast enough for offline scoring without a GPU.

The difference in training cost is larger than that in inference. In the seed-123 run, training and validating all ten folds took 33 h 36 min for E0 but 3 h 17 min for E3 and 2 h 45 min for E2, shorter by factors of 10.2 and 12.2 (Table 6). A full cross-validation of the ResNet-1D-TCN pipeline thus fits within a working session instead of taking more than a day, which shortens every development and verification cycle. The decomposition of amendment A2 shows where this difference arises. E0 trains 16 networks per fold, 15 CNN branches and a BiLSTM, whereas E3 trains a ResNet-1D encoder and a TCN. No CNN branch came close to its limit of 1,000 epochs: early stopping ended them after 29.1 epochs on average, against 19.0 epochs for the ResNet-1D encoders of E3 (limit 40). E0 therefore ran 22.5× as many optimiser steps as E3, while each of its steps took on average only 0.41 times as long, because one baseline CNN step (6.7 ms) is cheaper than one ResNet-1D step (16.1 ms). Steps multiplied by the measured time per step account for 59.5% of the wall-clock time of E0 and 66.1% of that of E3; the remainder covers validation, feature extraction, data loading and checkpointing (Supplementary Table S53). The ratio thus reflects how many networks each pipeline trains, not faster computation per step, and should not be attributed to the TCN or to a per-step efficiency of the ResNet-1D encoder.

Table 6. Training-and-validation wall-clock time over ten folds (seed 123; all configurations trained sequentially on the same Tesla V100). Ratios are E0 time divided by configuration time. E1 reuses the E0 encoder and cached features, so its time covers sequence-model training only. Optimiser steps are summed over all training stages and folds (amendment A2).

ID	Ten folds	Mean per fold	Optimiser steps	Faster than E0
E0	33 h 36 min	3 h 22 min	10,692,557	1.0×
E1 (sequence only)	14 min	1 min 26 s	n/a	n/a
E2	2 h 45 min	16 min 30 s	401,776	12.2×
E3	3 h 17 min	19 min 40 s	474,809	10.2×
E4	2 h 56 min	17 min 36 s	425,612	11.5×
E6	2 h 25 min	14 min 28 s	344,928	13.9×

Representation and context analyses, including a Sleep-EDF context ablation in which the previous- and next-epoch feature groups gave no detectable gain (Δmacro-F1 = 0.0010, $CI_{95\%}$ $[-0.0046, 0.0066]$), are reported in Supplementary Fig. S1 and Table S22.

4.3. Cross-cohort evaluation on the first SHHS1 sample

On the first SHHS1 sample, both primary comparisons favoured E3 (Table 7; descriptive performance in Supplementary Table S5), with improvements in 138 and 125 of 180 participants, and the seed-123 repeat preserved their direction and Holm significance. Relative to E0, E3 also gained 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461 κ. Among the secondary comparisons, E1−E0 was small (+0.0065) and E2−E1 negative (−0.0128, interval below zero), so replacing the C/P/N features by current-epoch ResNet-1D features did not improve transfer on unfiltered signals. The post-hoc E3−E2 contrast, which changes only the input filtering and constant scale, was the largest (+0.0475, $CI_{95\%}$ $[0.0372, 0.0579]$; 147 of 180 participants improved). Band-pass-only E4 scored higher than E3 at both seeds, by 0.0131 ($CI_{95\%}$ $[0.0111, 0.0153]$) and 0.0098 ($CI_{95\%}$ $[0.0073, 0.0125]$); Section 4.4 relates this difference to training variability (Supplementary Tables S13–S15).

Table 7. Primary paired comparisons on the first SHHS1 sample (subject-mean macro-F1, $n = 180$, seed 42).

Comparison	Role	Δ	$CI_{95\%}$	p_{Holm}	W/T/L	Δ_{sign}
E3−E0	primary	0.0412	[0.0314, 0.0512]	2.65×10^{-13}	138/0/42	+0.533
E3−E6	primary	0.0274	[0.0182, 0.0370]	1.87×10^{-8}	125/0/55	+0.389

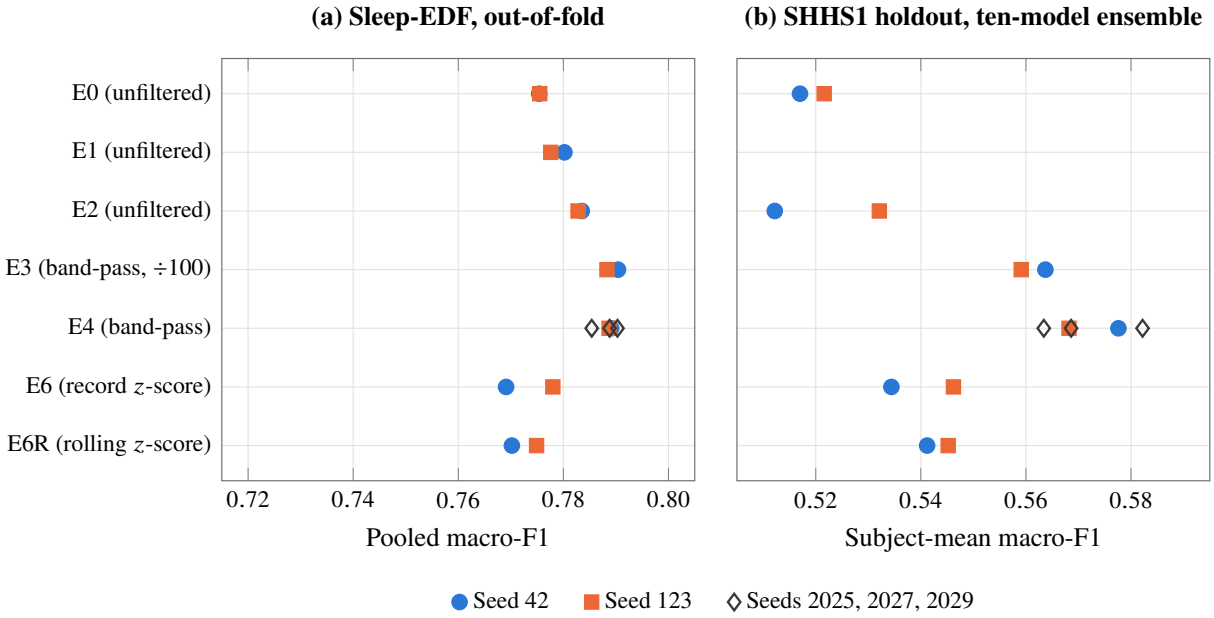


Figure 2. Scores of every configuration and training seed. (a) Pooled out-of-fold macro-F1 on Sleep-EDF. (b) Subject-mean macro-F1 of the ten-model ensemble on the holdout SHHS1 sample ($n = 500$). Each marker is one complete ten-fold training run; E4 has five seeds, and E1 was not evaluated on the holdout sample. Both horizontal axes span 0.09, so distances can be compared between panels although the two metrics are aggregated differently. E6R and the three further E4 seeds belong to amendment A2. Values are taken from Supplementary Tables S10, S38 and S47.

4.4. Confirmatory evaluation on the holdout SHHS1 sample

Table 8 reports the performance of each configuration on the holdout sample and Table 9 the confirmatory family. Fig. 2 shows the score of every configuration and training seed in both cohorts, and Fig. 3 sets the confirmatory contrasts against the difference between two training runs of the same configuration.

All three pre-specified hypotheses were confirmed (Table 9, Fig. 3a), and the holdout estimates lie close to those of the first sample. E3 improved on the comparator in 380, 342 and 395 of 500 participants; its subject-mean macro-F1 was 0.5637 on the holdout sample, compared with 0.5680 on the first sample. At seed 123, all three contrasts remained positive with intervals above zero. Results for all twelve arms and the secondary analyses are reported in Supplementary Tables S38–S43.

Thirty-nine holdout participants lacked at least one reference class, most often N3 (34 participants). Averaging macro-F1 only over the classes present in each participant's reference labels changed the three confirmatory estimates to 0.0482, 0.0296 and 0.0535, with the same decisions. Pooling the first and holdout samples (680 participants; descriptive only) gave 0.0452, 0.0288 and 0.0504.

The unfiltered ResNet-1D-TCN (E2) scored close to the baseline: subject-mean macro-F1 0.5122 versus 0.5170 for E0 at seed 42, and 0.5321 versus 0.5216 at seed 123. Most of E3's advantage over E0 therefore coincided with the change of input filtering rather than with the change of encoder and sequence model. This comparison of E2 with E0 is descriptive and was not part of the confirmatory family.

Training variability was of the same order as some configuration contrasts (Fig. 3b). Subject-mean macro-F1 at seed 42 minus that at seed 123 was +0.0046 ($CI_{95\%}$ [0.0026, 0.0067]) for E3, +0.0094

Table 8. Holdout SHHS1 sample (n = 500; 458,409 scored epochs): performance of the seed-42 configurations and the ADAST pair. Intervals are subject-cluster bootstrap intervals for subject-mean macro-F1. Source-only is the matched ADAST control, not E3.

ID	Subject-mean macro-F1 ($CI_{95\%}$)	Pooled macro-F1	Accuracy	κ
E0	0.5170 [0.5060, 0.5281]	0.5612	0.6506	0.5129
E2	0.5122 [0.5001, 0.5241]	0.5533	0.6512	0.5127
E3	0.5637 [0.5523, 0.5753]	0.6030	0.6957	0.5710
E4	0.5776 [0.5662, 0.5893]	0.6187	0.7103	0.5896
E6	0.5344 [0.5231, 0.5460]	0.5711	0.6740	0.5389
Source-only (full source)	0.4986 [0.4901, 0.5072]	0.5416	0.6844	0.5403
ADAST (full source)	0.5415 [0.5310, 0.5521]	0.5852	0.6798	0.5625

Table 9. Confirmatory family on the holdout SHHS1 sample (subject-mean macro-F1, n = 500). H1–H3 use seed 42 and a Holm family of three; seed 123 forms a separate, secondary Holm family. A hypothesis is confirmed if $p_{\text{Holm}} < 0.05$ and the interval lies above zero. The last column repeats the estimate from the first SHHS1 sample.

	Comparison	Δ	$CI_{95\%}$	p_{Holm}	W/T/L	Confirmed	First sample
H1	E3−E0	0.0467	[0.0403, 0.0531]	2.0×10^{-37}	380/0/120	yes	0.0412
H2	E3−E6	0.0293	[0.0231, 0.0354]	2.8×10^{-22}	342/0/158	yes	0.0274
H3	E3−E2	0.0515	[0.0460, 0.0571]	1.2×10^{-50}	395/1/104	yes	0.0475
Seed 123 (secondary)							
	E3−E0	0.0375	[0.0307, 0.0444]	1.1×10^{-22}	345/0/155	n/a	n/a
	E3−E6	0.0129	[0.0057, 0.0199]	1.7×10^{-5}	285/0/215	n/a	n/a
	E3−E2	0.0270	[0.0217, 0.0325]	1.8×10^{-18}	327/1/172	n/a	n/a

([0.0076, 0.0112]) for E4, −0.0046 ([−0.0063, −0.0029]) for E0 and −0.0118 ([−0.0151, −0.0085]) for E6; the unfiltered E2, not part of this pre-specified set, differed by −0.0199. All four intervals exclude zero even though each pair shares its configuration: intervals computed for fixed models do not capture variation between training runs. E4 again scored higher than E3, by 0.0139 ([0.0126, 0.0153]) at seed 42 and 0.0091 ([0.0074, 0.0108]) at seed 123. The pre-specified expectation that E4−E3 would not exceed the between-seed differences was met at seed 123 but not at seed 42, where E4−E3 was larger than each of the four pre-specified between-seed differences. With the five seeds of amendment A2 (Section 4.5), the between-seed differences of E4 reached 0.0189, so that both values of E4−E3 lie within the training variability of E4.

The trained weights are consistent with approximate input-scale invariance (Supplementary Table S44). For the same seed and fold, the first-layer kernels of E3 and E4 had cosine similarity 0.909–0.974 and the median similarity over encoder weight tensors was 0.696–0.863, against values between −0.112 and 0.012 for two seeds of the same configuration. The first-layer running variance of E4 was about 10^4 times that of E3, as the factor of 100 predicts, and early stopping selected different epochs for E3 and E4 in 8 of the 20 seed-fold pairs. The input-scale change was thus largely absorbed, and E4−E3 lay within the observed five-seed range of E4. By contrast, H1 and H3 exceeded every reported between-seed difference at both seeds, whereas H2 at seed 123 (0.0129) was close to the between-seed differences of E4 and E6.

4.5. Input spectra, normalisation scope and training variability

The first analysis below is a post-hoc description of the inputs; the other two belong to the second amendment (Section 3.10) and are descriptive.

Spectral content of the inputs. The input spectra show what the filter removes (Fig. 4; Supplementary Section S7). In the unfiltered signal, a median of 26.4% of the power of a Sleep-EDF recording (interquartile range 21.4–33.2%) and 21.0% of that of a holdout SHHS1 recording (14.4–28.9%) lay below 0.5 Hz, and 3.1% and 1.1% above 30 Hz. The cohorts differed most outside the pass-band: the median SHHS1 spectrum

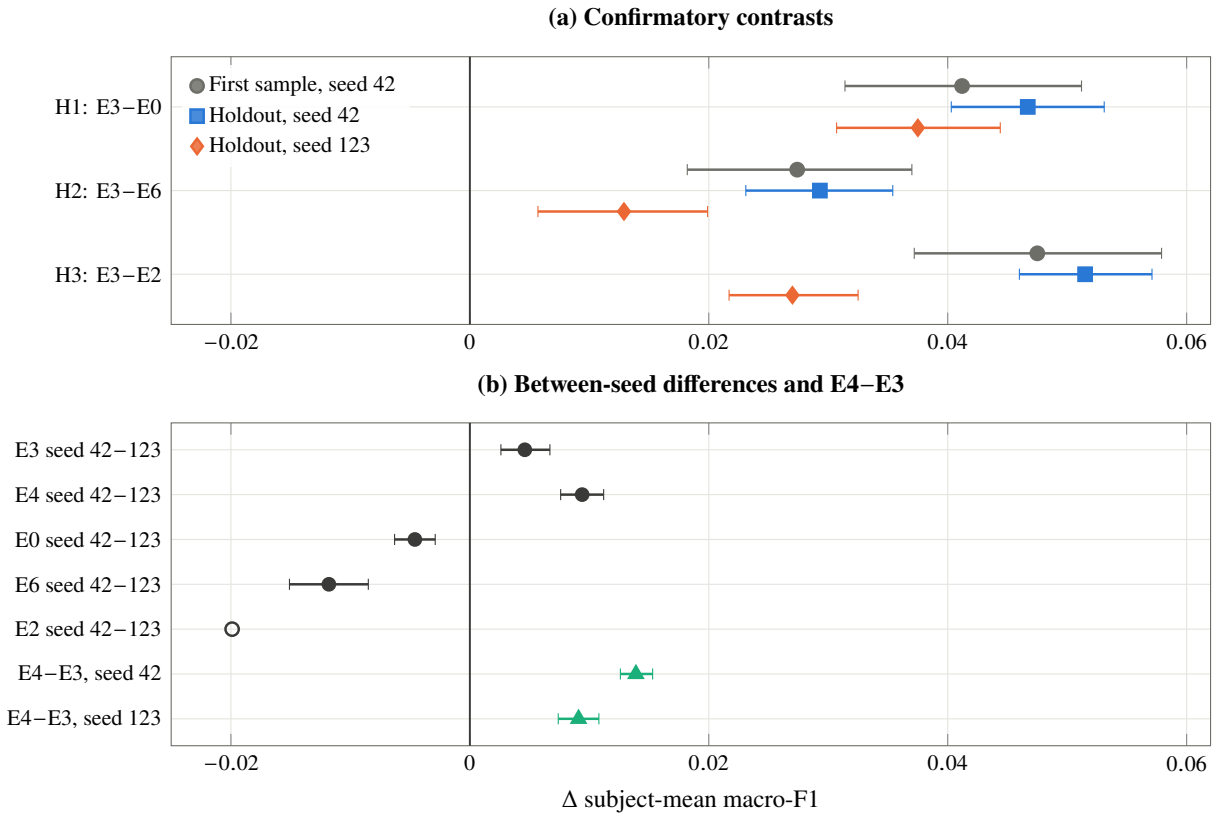


Figure 3. Confirmatory contrasts against training variability. (a) Differences in subject-mean macro-F1 for H1–H3 on the first SHHS1 sample (n = 180, seed 42) and on the holdout sample (n = 500, seeds 42 and 123). (b) Differences between two training seeds of the same configuration (seed 42 minus seed 123) and E4−E3 at each seed on the holdout sample; E2 has no pre-specified interval and is shown as a point estimate (open circle). Bars are 95% subject-cluster bootstrap intervals; both panels share the horizontal axis. Values are taken from Tables 9 and 7 and Supplementary Tables S16 and S40.

was 3.2–8.8 dB above that of Sleep-EDF between 0.1 and 0.2 Hz and 7.7–19.6 dB below it above 40 Hz, against differences between −2.5 and +4.4 dB within 0.5–30 Hz. Band-pass filtering removed the out-of-band content in both cohorts, leaving 98.6% and 99.4% of the remaining power within the pass-band, and left the in-band difference unchanged. The filter therefore discards the frequency ranges in which the unfiltered recordings of the two cohorts differ most.

Normalisation scope. Within the evaluation windows, the median per-recording standard deviation of the E6R input was 0.987 on Sleep-EDF and 0.993 on the holdout sample, against 0.770 and 0.992 for E6, so the scale mismatch of E6 was removed. Its effect on transfer was small: E6R minus E6 was 0.0068 ($CI_{95\%}$ [0.0012, 0.0124]) in subject-mean macro-F1 at seed 42 and −0.0010 ([−0.0071, 0.0051]) at seed 123, and E6R reached subject-mean macro-F1 of 0.5412 and 0.5452. E4 scored higher than E6R at both seeds, by 0.0364 ([0.0313, 0.0415]) and 0.0230 ([0.0177, 0.0282]), as did E3, by 0.0225 ([0.0171, 0.0278]) and 0.0139 ([0.0083, 0.0194]). By the locked rule, the statement that preserving calibrated amplitude transferred better than z-scoring is therefore retained. Windowed normalisation also lowered N3 recall below that of E6, to 0.1395 and 0.1515 (differences −0.0591 and −0.0503), with N3→N2 rates of 84.2% and 82.8%, whereas E4 exceeded E6R in N3 recall by 0.1346 and 0.1059. On Sleep-EDF, E6R reached a pooled macro-F1 of 0.7702 and 0.7749, close to E6 (0.7691 and 0.7780) and below E4 (0.7891 and 0.7887).

Training seeds. Across the five seeds of E4, holdout subject-mean macro-F1 had a mean of 0.5720, a standard deviation of 0.0077 and a range of 0.5634–0.5822, whereas pooled Sleep-EDF macro-F1 varied with a standard deviation of 0.0018 (range 0.7854–0.7903); the two dispersions use different aggregations (subject-mean ensemble scores versus pooled out-of-fold scores) and are not directly comparable. Nine of the ten pairwise between-seed intervals excluded zero, and the largest absolute difference was 0.0189. All 20

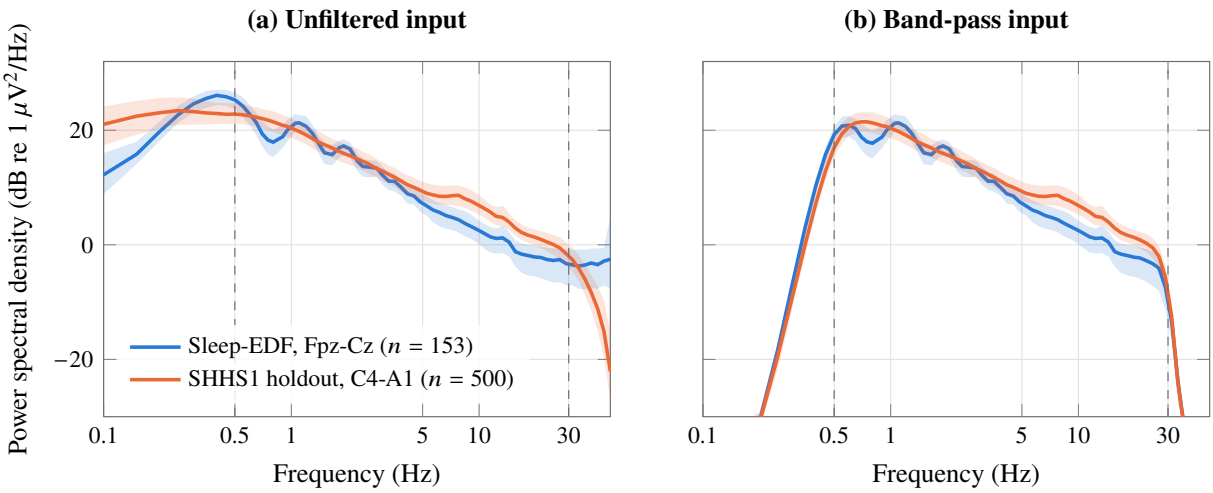


Figure 4. Power spectra of the model inputs in both cohorts. Median over recordings (line) and interquartile range (shading) of the Welch power spectral density within the evaluation window, for (a) the unfiltered input of E0–E2 and (b) the band-pass input of E4. Dashed lines mark the 0.5–30 Hz pass-band. Post-hoc descriptive analysis; no model is involved. Values are taken from Supplementary Section S7.

differences between an E4 seed and E0 or E2 at either seed were larger, 0.0313–0.0700 with every interval above 0.026, so by the locked rule these configuration contrasts exceed the observed between-seed range of E4 (Supplementary Tables S47–S50).

4.6. Cross-model errors and their temporal distribution

E3's pooled macro-F1 fell from 0.7904 on Sleep-EDF to 0.6099 on the first SHHS1 sample, a difference of 0.1805. Because the two cohorts are evaluated differently, by out-of-fold predictions and by ten-model ensembles, we examine this difference descriptively by class. On the holdout sample, the corresponding difference was 0.1875 with the ensemble and 0.2107 when the fold models were scored individually, as on Sleep-EDF (below). A→B denotes a reference-labelled A epoch predicted as B. N3 accounts for most of the loss (Fig. 5): E3 N3 F1 fell from 0.8085 to 0.4055, with precision 0.9440 but recall 0.2582, and E0 showed nearly the same recall (0.2610) and N3→N2 rate (72.3% versus 73.1%; Supplementary Table S6).

The holdout sample reproduced this pattern: N3→N2 rates were 72.3% for E0 ($CI_{95\%}$ 70.1–74.4%), 72.7% for E3 (70.7–74.7%), 71.1% for E4 and 77.1% for E6, with N3 recall of 0.2565, 0.2549, 0.2742 and 0.1987 and N3 precision between 0.85 and 0.88, each with an interval about 0.10 wide, against 0.90 to 0.94 on the first sample.

In an oracle calculation on E3's first-sample confusion matrix, correcting the N3→N2 cell would raise pooled macro-F1 to 0.7444, 74.5% of the cross-cohort difference, against 0.6596 (27.5%) for N2→REM, the second-largest channel; these reference-based figures are neither additive nor achievable. N3 recall was low not only near reference-label changes (0.0633 for E0 and 0.0733 for E3, against 0.6121 and 0.6326 on Sleep-EDF) but also in stable regions (0.4008 and 0.3900), and record-wise z-scoring did not remove the deficit (E6 N3 recall 0.2005). Region definitions and all class-wise results are given in Supplementary Tables S17–S21.

Single models and ensembles. In the second amendment, every fold model was also scored alone. Single fold models of one arm differed by up to 0.18 in holdout subject-mean macro-F1, and the ten-model ensemble exceeded the mean of its single models by 0.0121–0.0327 in every arm, each interval above zero (Supplementary Table S51). N3 recall was slightly higher for single models than for ensembles (for E3 at seed

the drop in pooled macro-F1 from Sleep-EDF to the holdout sample was 0.2107 and 0.2134 for E3 at seeds 42 and 123, compared with 0.1875 and 0.1902 with the ensembles; the cohort comparison above therefore understates the cross-cohort loss of a single model by about 0.02.

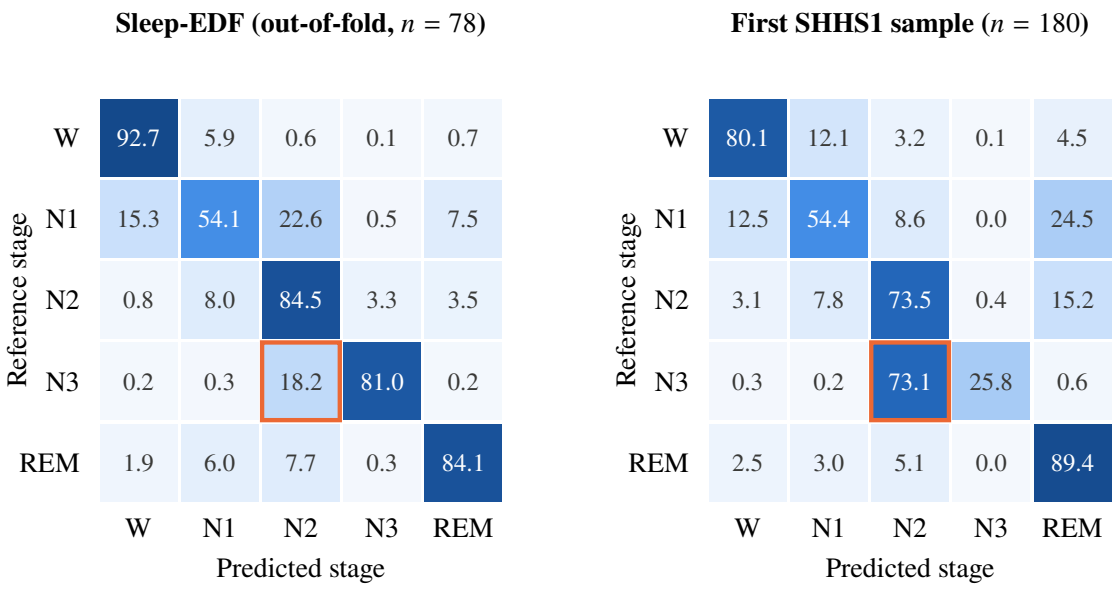


Figure 5. E3 confusion matrices (seed 42) before and after cross-cohort transfer. Each cell is the percentage of epochs of a reference stage (row) predicted as each stage (column); rows sum to 100%. The orange outline marks N3→N2, which rises from 18.2% on Sleep-EDF to 73.1% on the first SHHS1 sample. Counts are taken from Supplementary Tables S9 and S18.

4.7. Intervention results

On the first SHHS1 sample, temperature scaling barely changed subject-mean macro-F1 (0.5590 to 0.5595), whereas adding EM prior correction reduced it to 0.4899 (difference −0.0696, $CI_{95\%}$ [−0.0793, −0.0600]): the fitted priors placed N3 near the floor, and the ensemble predicted no N3 epochs (Supplementary Table S23). Class weighting raised N3 recall from 0.2345 to 0.3225 (difference 0.0880, $CI_{95\%}$ [0.0805, 0.0954]) and N3 F1 from 0.3759 to 0.4795, but lowered N2 recall from 0.7260 to 0.6728 and accuracy from 0.6892 to 0.6663, leaving subject-mean macro-F1 unchanged (difference −0.0016, $CI_{95\%}$ [−0.0049, 0.0017]); this intervention acts on the sequence classifier only (Supplementary Table S24).

At the limited budget, ADAST raised N3 recall from 0.3195 to 0.4269 but lowered subject-mean macro-F1 from 0.4949 to 0.4497 (difference −0.0452, $CI_{95\%}$ [−0.0558, −0.0347]), produced no N1 predictions and also lowered source performance. At the full-source budget the effect reversed: subject-mean macro-F1 rose from 0.5010 to 0.5455 (difference 0.0445, $CI_{95\%}$ [0.0352, 0.0539]; difference between the budget effects 0.0897, $CI_{95\%}$ [0.0767, 0.1023]) and N3 recall from 0.2432 to 0.6111, whereas N1 and N2 recall fell and higher REM recall came with lower REM precision, reducing REM F1 from 0.6231 to 0.5744. Source outer-test subject-mean macro-F1 fell from 0.6613 to 0.6281. All classes, checkpoint rules and budgets are reported in Supplementary Tables S7, S26, S36 and S37.

The holdout sample reproduced the full-source result. ADAST exceeded its control by 0.0429 in subject-mean macro-F1 ($CI_{95\%}$ [0.0374, 0.0484]) and raised N3 recall from 0.2366 to 0.6161, while N1, N2 and REM F1 decreased by 0.0190, 0.0445 and 0.0568. In absolute terms, however, ADAST remained below the zero-shot band-pass pipelines: its subject-mean macro-F1 was 0.0222 lower than that of E3 ([−0.0284, −0.0160]) and 0.0361 lower than that of E4 ([−0.0423, −0.0299]). Both ADAST arms used the E3 input, so this gap mainly reflects the ADAST backbone and its training rather than the adaptation: without adaptation, the same backbone (the matched source-only control) scored 0.4986, 0.0651 below E3 (Table 8), and adaptation recovered 0.0429 of that difference.

Sample size of the EM prior estimate. The second amendment repeated the prior correction on the holdout sample with priors estimated from samples of increasing size. There, the archived priors from the five adaptation recordings again placed N3 at the floor, and the adjusted ensemble predicted no N3. Estimating the priors from larger samples did not resolve the problem: with the 20,000 epochs of the AdaBN

sample or with all 500 recordings, the fold-mean N3 prior was 0.030 and 0.029, about a fifth of the reference N3 share of 0.134, and N3 recall fell from 0.240 without adjustment to 0.134 and 0.129, with subject-mean macro-F1 of 0.517 and 0.516 against 0.559. Random subsets of 5, 20 and 100 recordings converged on the same prior (median 0.028–0.029) as their spread narrowed (Supplementary Table S52). By the locked rule, the collapse is therefore attributed to the label-shift assumption failing under feature shift, not to the small adaptation sample.

4.8. Amplitude sensitivity and batch-normalisation adaptation

The band-pass-filtered SHHS1 signal was not smaller than the Sleep-EDF signal: its per-recording standard deviation had a median of 16.50 μV, interquartile range 13.72–21.26, on the holdout sample and of 14.36 μV, interquartile range 11.63–16.82, on Sleep-EDF.

Nor had reference N3 epochs lost their amplitude contrast with N2 (Supplementary Section S8). The per-recording median standard deviation of N3 epochs was 20.8 μV (interquartile range 16.3–23.3) on Sleep-EDF and 18.8 μV (16.9–21.1) on the holdout sample, against 11.5 μV (9.4–13.7) and 13.0 μV (11.7–14.7) for N2. Within a recording, N3 epochs had 1.62 (1.44–1.80) and 1.43 (1.34–1.54) times the standard deviation of N2 epochs and 4.9 and 4.2 dB more power between 0.5 and 4 Hz. The contrast was therefore smaller in SHHS1 but still large. Epoch standard deviation alone separated reference N3 from reference N2 epochs with an AUC of 0.940 on Sleep-EDF and 0.892 on the holdout sample, and the threshold that maximised balanced accuracy on Sleep-EDF (16.5 μV) placed 84.0% of the 61,523 holdout N3 epochs above it and 82.2% of the holdout N2 epochs below it (77.4% and 88.8% for 0.5–4 Hz power). This two-class description is not a five-stage classifier, but it shows that the amplitude difference between N3 and N2 on which a Sleep-EDF threshold relies is present in most SHHS1 N3 epochs, of which the models recognised about a quarter.

Input amplitude nevertheless strongly influenced the separation of N2 from N3 (Fig. 6; full values in Supplementary Tables S45 and S46). As the input factor increased from 0.5 to 2, the proportion of reference N3 epochs predicted as N2 fell monotonically from 93.3% to 48.6%. N3 recall, however, rose only up to a factor of 1.5, from 0.2742 to 0.3559, a difference of 0.0817 with $\mathrm{CI}_{95\%}$ $[0.0665, 0.0974]$. At a factor of 2 it fell to 0.2318, because 28.2% of reference N3 epochs were then classified as wake, and REM recall dropped from 0.876 to 0.276. At a factor of 1.5, the gain in N3 F1 of 0.0859, $\mathrm{CI}_{95\%}$ $[0.0697, 0.1020]$, was offset by lower N1, REM and wake F1, and subject-mean macro-F1 was unchanged (-0.0011, $\mathrm{CI}_{95\%}$ $[-0.0099, 0.0075]$). Factors below 1 reduced N3 recall to 0.1327 and 0.0176. The response was thus monotonic for the N3→N2 rate but not for N3 recall, which by the pre-specified criterion argues against an amplitude-only explanation of N3 under-detection.

AdaBN changed subject-mean macro-F1 by -0.1206, $\mathrm{CI}_{95\%}$ $[-0.1295, -0.1116]$, without a detectable change in N3 recall (difference 0.0086, $\mathrm{CI}_{95\%}$ $[-0.0052, 0.0222]$). Wake precision fell from 0.853 to 0.589 and REM recall from 0.876 to 0.312. AdaBN also scored below ADAST, by 0.0844 with $\mathrm{CI}_{95\%}$ $[0.0759, 0.0929]$, and below the matched source-only control of ADAST, by 0.0416 with $\mathrm{CI}_{95\%}$ $[0.0333, 0.0498]$. Supplementary Section S5 reports all classes and contrasts.

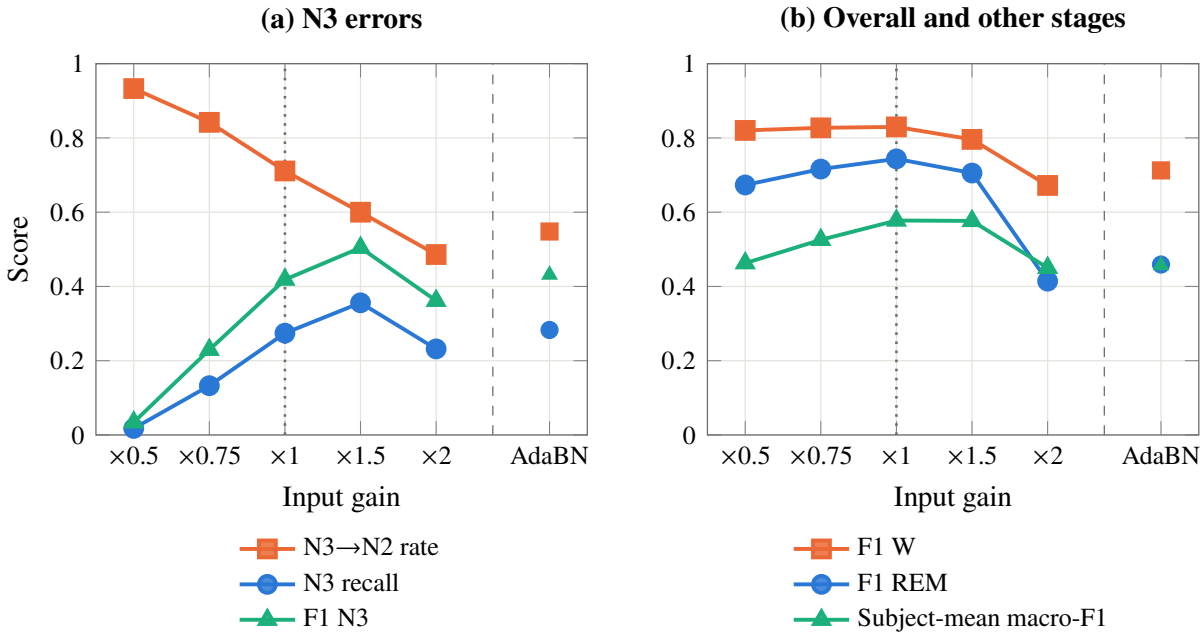


Figure 6. Amplitude sensitivity and AdaBN for E4 (seed 42) on the holdout SHHS1 sample (n = 500)**.** The filtered input was multiplied by 0.5–2 while the encoder kept its stored batch-normalisation statistics; ×1 (dotted line) is unmodified E4. AdaBN replaces the batch-normalisation statistics with those of unlabelled SHHS1 epochs and is shown separately, right of the dashed line. (a) The N3→N2 rate falls monotonically as amplitude increases, but N3 recall rises only up to ×1.5. (b) Subject-mean macro-F1 does not exceed that of E4 at any factor, and wake and REM F1 drop sharply at ×2 and with AdaBN. Class-wise metrics are pooled over epochs; descriptive analysis without hypothesis tests. Values are taken from Supplementary Table S45.

5. Discussion

5.1. What the pipeline comparisons establish

The confirmatory evaluation supports three conclusions about the evaluated pipelines. First, the ResNet-1D-TCN configuration on band-pass-filtered input transferred better to SHHS1 than the CNN-BiLSTM baseline, and this advantage, first observed on 180 participants, was confirmed on 500 participants whose data had not informed any analysis. It was also much cheaper to develop and use: ten-fold training and validation took about 3.3 h instead of 33.6 h on the same GPU, and inference was 3.76× faster in the GPU benchmark and 3.1× faster for a whole night on a CPU. For a pipeline that must be retrained for every preprocessing variant, seed and fold, this reduction in training time is a practical advantage. It arises because the baseline trains 15 encoder networks per fold, each stopped early after about 30 epochs, and therefore needs 22.5× as many optimiser steps, not because a ResNet-1D step is cheaper; a baseline CNN step in fact takes less time. The ratio therefore describes the two pipelines as specified, not the per-step efficiency of the architectures.

Second, band-pass filtering accounted for most of this advantage. The same model on unfiltered input performed close to the baseline on the holdout sample, and on the first sample neither the sequence-model replacement (E1−E0) nor the feature/context replacement (E2−E1) produced a comparable gain. How the signal is filtered is therefore a substantial part of how these models transfer, not an implementation detail.

Third, preserving calibrated amplitude (E3) transferred better than record-wise z-scoring (E6). Because the statistics of E6 span about 23 h of mostly daytime recording in Sleep-EDF but only the night in SHHS1, the two cohorts reach E6 at different relative scales, so E6 alone is not a clean test of normalisation. Matching the span with a 90-minute window (E6R) removed this mismatch but hardly changed transfer, and both band-pass configurations still scored above E6R at both seeds. The deficit of z-scoring therefore does not stem from the mismatched span; the lower N3 recall of E6R suggests instead that removing absolute amplitude discards information the model uses to recognise N3. E6R also removes amplitude changes slower than its window, so this comparison concerns these two forms of z-scoring, not every possible normalisation.

These are comparisons of whole configurations, including their training recipes. The confirmed contrasts establish the effect of the filtering step on transfer, and the input spectra indicate where it may arise: the unfiltered recordings of the two cohorts differ most below 0.5 Hz and above 30 Hz, the ranges the filter removes, whereas their in-band spectra differ by a few decibels with or without filtering. The comparisons concern the complete 0.5–30 Hz filter and do not separate its two edges.

5.2. Training variability and the E3−E4 contrast

On the holdout sample, two training seeds of an identical configuration differed by 0.005 to 0.012 in subject-mean macro-F1 for the four pre-specified configurations and by 0.020 for E2, and all four pre-specified between-seed intervals excluded zero. Five seeds of E4 spread over 0.019 (standard deviation 0.0077), and nine of their ten pairwise intervals excluded zero. These intervals quantify participant-sampling uncertainty for fitted models and do not include variation from training new models, so a precisely estimated difference between two fitted models can still be of the size observed between training seeds.

E4−E3 illustrates this. The two variants differ by a fixed input-scale factor that batch normalisation largely absorbs, and their checkpoints are far more similar to each other than to models trained with another seed; finite ε, running-statistic initialisation and numerical precision nevertheless let their training diverge. Their SHHS1 subject-mean macro-F1 differences of 0.009–0.014 lie within the five-seed range of E4. H2 at seed 123 (0.0129) was likewise comparable to some between-seed differences, so its participant-level significance should be read together with its sensitivity to retraining.

Every difference between an E4 seed and E0 or E2 (0.031–0.070) exceeded the largest between-seed difference of E4, so the advantage of the band-pass pipeline over these comparators was consistent across the evaluated seeds, although five and two seeds do not characterise the full distribution of retraining effects. Reporting the training spread alongside participant-level intervals makes both sources of uncertainty explicit.

For practical use, we recommend band-pass-only E4 as the simpler evaluated option: clipping never activated in these data, and E4 transferred well without the fixed scaling step. E3 remains the reference for the confirmatory hypotheses because the protocol fixed it before the holdout data were processed; the E4 and seed analyses support, but do not replace, those contrasts.

5.3. N3 under-detection persists alongside aggregate improvement

Despite E3's overall advantage, the class-wise analysis shows an error shared by all models. E0 and E3 both classified more than 72% of reference N3 epochs as N2 on both SHHS1 samples, with N3 recall near 0.26 despite precision above 0.85; band-pass-only E4 and record-wise z-scored E6 showed the same pattern. The oracle calculation shows the numerical importance of N3→N2 for E3, and the boundary analysis shows that the deficit extends beyond the neighbourhoods of reference-label changes into stable N3. E3's better aggregate transfer therefore left the largest stage-specific error in place. The deficit is not specific to the present implementation: the original ZleepAnlystNet models trained on Sleep-EDF Fpz-Cz reached an N3 F1 of only 0.43 on 200 SHHS recordings from the C3-A2 channel, a different derivation from the one used here [2].

The cohorts differ simultaneously in EEG derivation, population, acquisition and scoring conditions. Slow-wave spatial structure and amplitude-based N3 scoring criteria motivate examining these differences [31, 34]. The amplitude sweep examined one of these factors directly. The filtered SHHS1 signal was not smaller than the Sleep-EDF signal, yet the model's N2/N3 decision depended strongly on input amplitude: enlarging the input moved reference N3 epochs from N2 towards N3 and, beyond a factor of 1.5, towards wake. This behaviour is consistent with a decision boundary that, like the scoring rules, relies on slow-wave amplitude. However, no single factor recovered N3 without degrading other stages, and the most favourable factor still left 60% of N3 epochs classified as N2. A global amplitude mismatch is therefore not a sufficient explanation. Nor is a loss of slow-wave amplitude at C4-A1: with a threshold chosen on Sleep-EDF, epoch amplitude alone placed 84% of SHHS1 N3 epochs on the N3 side, more than three times the N3 recall of the trained models. The amplitude cue that separates N3 from N2 is thus available in the target signal, and the models do not use it as a fixed threshold would. Which properties of Sleep-EDF N3 they rely on instead, and how far differences between scorers contribute, lies outside what the present design can separate.

Table 10. Published single-channel results on the 78-subject Sleep-EDF Expanded cassette recordings (Fpz-Cz) and the present configurations at seed 42. Published values are those of the original studies as compiled in [12], and those of ZleepAnlystNet are from [2]; evaluation protocols differ between studies.

Model	Accuracy (%)	macro-F1 (%)	κ
SleepEEGNet [41]	80.0	73.6	0.73
SleepTransformer [7]	81.4	74.3	0.743
SeqSleepNet [8]	82.6	76.4	0.760
TinySleepNet [9]	83.1	78.1	0.77
XSleepNet [10]	84.0	77.9	0.778
ZleepAnlystNet [2]	84.1	78.3	0.779
SleePyCo [12]	84.6	79.0	0.787
E0 (CNN-BiLSTM, unfiltered)	82.6	77.5	0.761
E3 (ResNet-1D-TCN, band-pass, fixed scale)	83.6	79.0	0.775
E4 (ResNet-1D-TCN, band-pass only)	83.6	78.9	0.775

The interventions changed the N3 error in different ways. EM prior correction eliminated N3 predictions. This was not a consequence of estimating the priors from only five recordings: with priors from all 500 holdout recordings, EM still estimated an N3 share of 0.029 against 0.134 and lowered N3 recall. Because the models under-detect N3 after transfer, their outputs understate N3, and EM, which assumes that only the class proportions differ between cohorts, propagates this bias into the priors; under feature shift the label-shift assumption does not hold. Class weighting raised N3 recall at the expense of N2 recall and accuracy. ADAST changed sign between learning budgets; with full-source training it more than doubled N3 recall and improved on its matched control on both SHHS1 samples, but lowered N1, N2 and REM F1 and source-cohort performance. ADAST was evaluated as published, with its own backbone, so its effect is the gain over its matched control; its absolute score below zero-shot E3 and E4 reflects the different backbones and is not interpreted as an effect of adaptation. AdaBN, applied to the E4 models themselves, left N3 recall unchanged and lowered aggregate performance by shifting the wake and REM boundaries. Cohort-level adaptation therefore either left the N3 error unchanged or traded it against other stages. The sign change between budgets also shows that the learning budget must be reported when an unsupervised domain-adaptation baseline is compared, and greater REM recall without greater REM F1 shows that recall gains must be read together with precision.

5.4. Positioning and limits of generalisation

This study combines paired pipeline comparisons, computational cost, an analysis of cross-cohort errors and a confirmatory evaluation on held-out participants. It complements work on input representations, conventional and deep pipelines, and multi-cohort training [14, 15, 17, 18]. E0 is a reimplementation of the ZleepAnlystNet design under the stated protocol and scored 0.0072 below the published Sleep-EDF macro-F1 of that design. On Sleep-EDF, the band-pass configurations lie within the range of published single-channel results (Table 10); because fold structure, validation split, wake trimming and model selection differ between studies, the table positions the present pipelines and is not a ranking. The primary comparisons evaluate transfer without weight updates. ADAST evaluates target-domain weight updates against a matched source-only control, and AdaBN evaluates target-domain normalisation statistics without weight updates.

The scope of the findings is as follows. The study evaluates transfer from one source cohort to one target cohort; the holdout sample is new in its participants but comes from the same SHHS1 study, montage and scoring procedures, so it confirms the findings within SHHS1. The confirmatory protocol was archived in the authors' public repository, and the GitHub server timestamps establish its existence before the analysis. The intervention experiments were designed after the errors of the first SHHS1 sample had been examined, and the amplitude, AdaBN and second-amendment analyses were specified after the confirmatory results; these analyses are descriptive. Five seeds describe the spread of training variability rather than the tails of

its distribution. Sleep-EDF is scored with out-of-fold predictions and SHHS1 with ten-model ensembles, so the cohort comparison combines cohort and evaluation-procedure differences, of which the ensembles account for about 0.02. The models use future context, so forward-pass timing describes offline processing. Class-wise results are pooled across participants without sex- or age-stratified analysis. Evaluation on further cohorts, montages and subgroups would show whether the filtering effect and the N3 pattern generalise.

6. Conclusion

Under a shared protocol, the ResNet-1D-TCN pipeline on band-pass-filtered EEG ran three to four times faster than the CNN-BiLSTM baseline, needed about one tenth of the training time because it trains two networks per fold instead of sixteen, and transferred better from Sleep-EDF to SHHS1. Its advantage, first observed on 180 participants, was confirmed on 500 participants held out from development: subject-mean macro-F1 improved by 0.0467 over the baseline, by 0.0293 over record-wise z-scoring and by 0.0515 over the same model on unfiltered input, and across five seeds of the band-pass-only variant and two seeds of each comparator, its contrasts with the baseline and the unfiltered model exceeded the observed between-seed range. Band-pass filtering accounted for most of the transfer gain and removed the frequency ranges in which the unfiltered recordings of the two cohorts differed most, and z-scoring remained inferior when its statistics covered the same span in both cohorts. Because clipping never activated and batch normalisation largely absorbs a constant input scale, band-pass filtering alone is the simpler form of the pipeline. All zero-shot pipelines classified most N3 epochs as N2. This decision depended strongly on input amplitude, yet most SHHS1 N3 epochs remained separable from N2 by an amplitude threshold chosen on Sleep-EDF, and none of rescaling, prior correction, class weighting, batch-normalisation adaptation or adversarial domain adaptation removed the error without losses in other stages. Pipeline comparisons for sleep staging should therefore report computational cost, stage-specific errors and variation between training runs, and should be confirmed on data that did not inform the comparisons.

Reproducibility and data availability

Code, protocols, data partitions and aggregate analyses are available in the project repository at https://github.com/quanuiter/SleepTCN. Sleep-EDF Expanded is available from PhysioNet (https://physionet.org/content/sleep-edfx/1.0.0/); access to SHHS and its conditions of use are managed by NSRR (https://sleepdata.org/datasets/shhs). Because access to the source records is controlled by the data provider, the repository and the manuscript package contain no participant-level SHHS records or predictions.

The confirmatory protocol, its analysis script and amendments A1 and A2 were each committed to the repository before they were run; their commit identifiers, and a timestamp recorded by the GitHub server independently of the authors' commit metadata, are listed in Supplementary Sections S3, S5 and S6. The holdout participants are selected by a deterministic hash ranking that can be reproduced from the NSRR participant list, and the inference manifests record the SHA-256 values of the protocol, the preprocessed inputs and the model checkpoints. Provenance checks for the first SHHS1 sample are given in the supplement and the repository artifact index.

Ethics

This study is a secondary analysis of de-identified repository data; no new participants were recruited. Access to SHHS data was obtained through the National Sleep Research Resource under its data use agreement. No additional ethics approval was required for this secondary analysis of de-identified repository data.

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CRediT authorship contribution statement

Ngo Nhat Quan: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing. Pham Thai Son: Validation, Investigation, Visualization, Writing – review and editing. Tri Nguyen Ho Duy: Supervision, Writing – review and editing. Tri Nguyen: Supervision, Writing – review and editing.

Declaration of competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used OpenAI Codex and Anthropic Claude. OpenAI Codex assisted with language editing, manuscript organisation, code and result consistency checks, reproducible plotting, preparation of the vector study-design schematic, and the implementation of analysis, inference and training scripts and their verification tests. Anthropic Claude assisted with code review, drafting of the confirmatory holdout protocol, implementation of the holdout preprocessing, inference and analysis scripts, the amendment, spectral, stage-wise and CPU-timing analyses and their tests, the plotting code for Figs. 2–6 from the locked values, and language editing of the manuscript. All numerical results were computed by these scripts from the recorded datasets and model checkpoints; none was supplied by a language model. After using these tools, the authors reviewed and edited the implementation, experimental design, results and text as needed and take full responsibility for the content of the publication.

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